Decentralization and Criminal Gangs in El Salvador: Impacts on Municipal Finances and Local Economic Development

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Introduction

Motivation

- Decentralization played a key role in El Salvador's post-civil war transition (1992 onward).
  - Municipalities became spaces of political contestation.
  - FMLN won local elections $\Rightarrow$ foundation for national victories.
  - Participatory mechanisms broadened decision making.

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  - FMLN won local elections ⇒ foundation for national victories.
  - Participatory mechanisms broadened decision making.
- More recently, a different non-state armed actor has undermined this progress: **criminal gangs (maras)**.
  - MS-13 and Barrio 18: up to 60,000 members.
  - Operated in nearly all 262 municipalities.
  - Widespread extortion of households and businesses.

This Paper

Key Question

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- Existing literature: decentralization $\Rightarrow$ leverage for gangs (Rios, 2013; Durán-Martínez, 2018).
- We **reverse the causal arrow**: gangs $\Rightarrow$ undermine decentralization.

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- Existing literature: decentralization ⇒ leverage for gangs (Rios, 2013; Durán-Martínez, 2018).
- We **reverse the causal arrow**: gangs ⇒ undermine decentralization.
- **Findings:**
  - Gang presence reduces own-source revenues (especially mid-size municipalities).
  - Gang presence reduces municipal spending and service provision.
  - Gang presence reduces local economic activity, with spillovers to neighboring municipalities.

Fit in the Literature

Gangs and the state: symbiosis or parasitism?

- Blattman et al. (2021): Gangs provide governance to protect drug rents $\Rightarrow$ complementarity with the state.
- Lessing (2021): Criminal governance as complement to state governance.
- Di Cataldo & Mastrorocco (2021): Organized crime distorts public service provision.

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Our contribution

In El Salvador, gangs do **not** rely on drug trafficking ⇒ no incentive to provide order. Instead, heavy reliance on **extortion** creates a **parasitic** relationship that hollows out municipal governance.

Mechanisms

Gangs disrupt the causal mechanisms behind decentralization's benefits:

Factor mobility	$\Rightarrow$	Checkpoints, extortion on transport of goods
Own-source revenue collection	$\Rightarrow$	Competing "taxation" by gangs
Service provision	$\Rightarrow$	Territorial control limits municipal operations
Local economic activity	$\Rightarrow$	Extortion raises costs, depresses production

Decentralization in El Salvador

Institutional Design

- Unitary state with **262 municipalities** (only subnational level).
- **Political**: Majoritarian electoral rules; since 2015, some opposition seats on council.
- **Fiscal**:
  - No municipal property tax.
  - Own-source revenues: fee-based taxes on economic activities, user fees, sale of goods/services.
  - Transfers: FODES (Fund for Municipal Economic and Social Development), since 1988.
  - FODES rule: 80% capital investment, 20% operating expenses.
- **Administrative**: Trash collection, local policing, housing, transport regulation, local economic development.

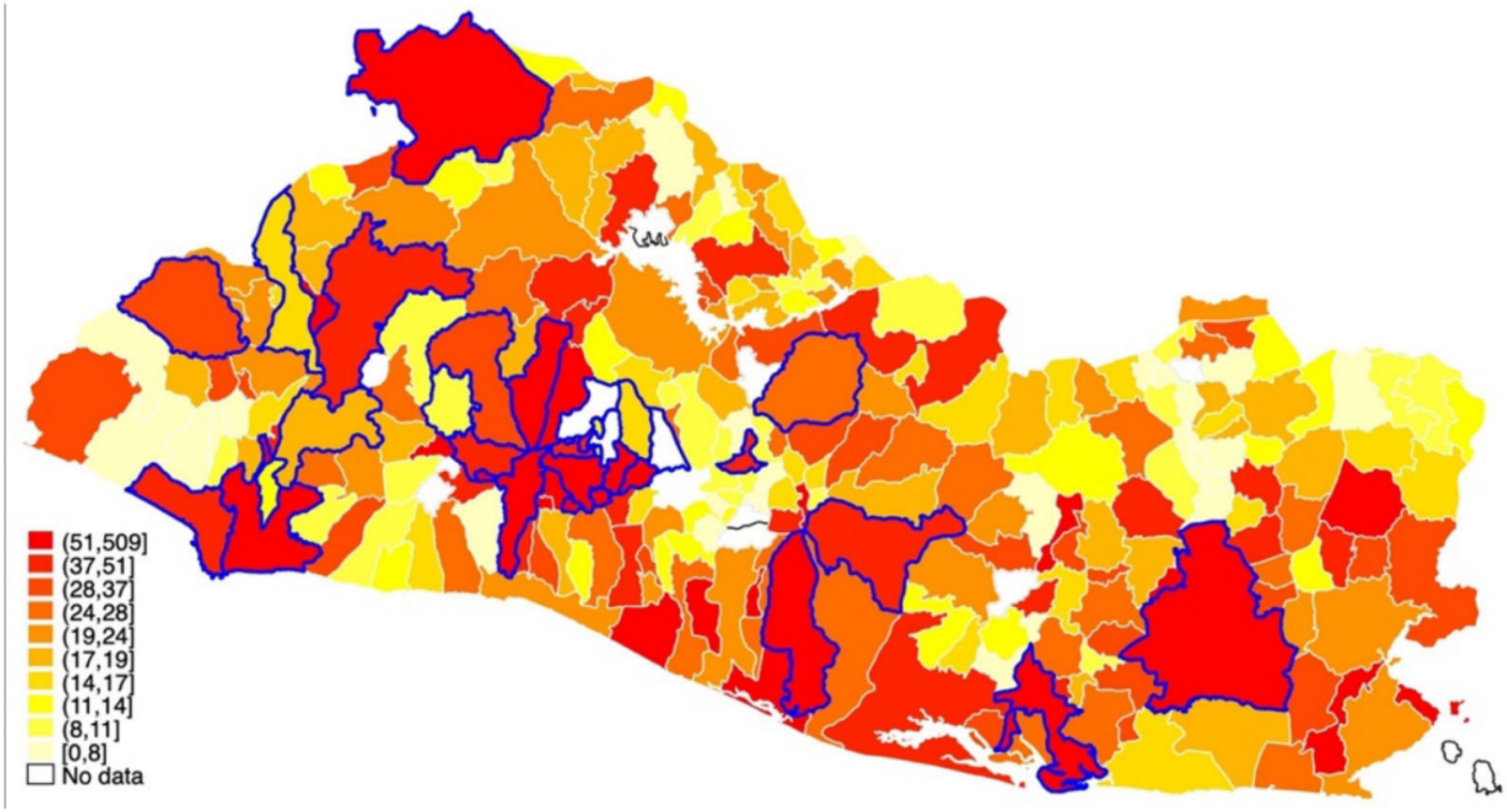
Municipal Finances: Key Facts (2010–2018)

- High reliance on transfers: FODES $\approx$ 48% of total municipal revenues.
- Low own-source revenues $\approx$ 37% of total revenues.
- Enormous heterogeneity by municipality size:
  - Small municipalities: FODES = 10$\times$ own-source revenues.
  - Medium: FODES = 3.5$\times$ own-source revenues.
  - Large: FODES = 1.2$\times$ own-source revenues.

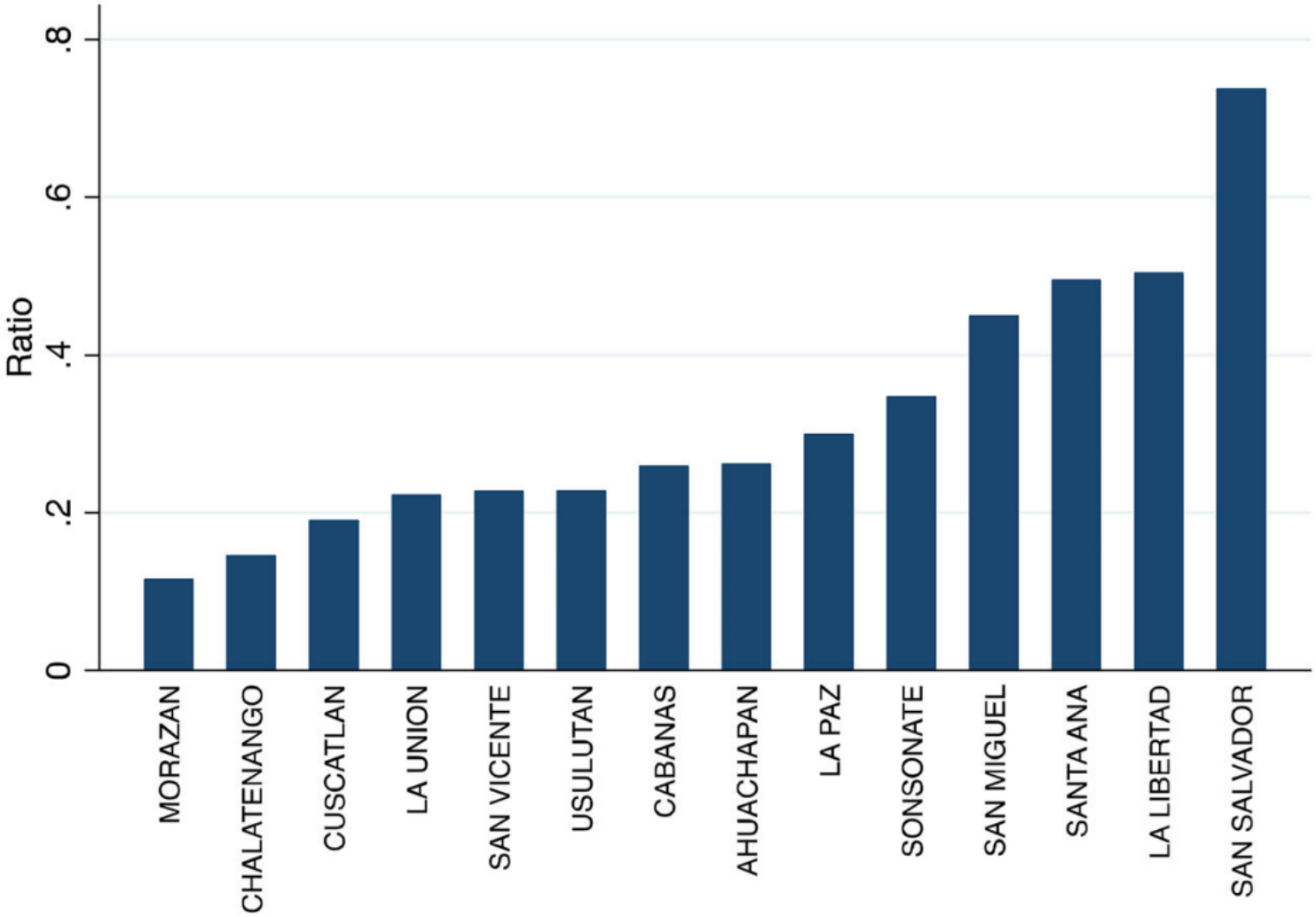
Per capita tax collection and size of FODES transfers, 2011–2018

	Small		Medium-sized		Large	
Year	Tax collection PC	Ratio of FODES own revenues	Tax collection PC	Ratio of FODES own revenues	Tax collection PC	Ratio of FODES own revenues
2011	13.8	25.0	27.7	4.2	35.1	1.2
2012	13.9	13.7	29.0	5.1	36.5	1.4
2013	15.7	12.4	31.9	4.7	39.0	1.3
2014	16.6	13.5	31.1	4.5	41.4	1.2
2015	17.1	14.5	33.0	4.5	43.9	1.2
2016	18.4	13.0	35.0	4.2	46.3	1.2
2017	19.2	11.1	34.2	3.8	46.2	1.1
2018	21.5	10.0	39.0	3.5	51.5	1.2

Own-Source Revenues Per Capita (2018)



Budgetary Autonomy by Department (2018)



Empirical Strategy

Empirical Approach

Panel data model with fixed effects:

- 258 municipalities, 2010–2019.

$$Y_{i,t} = \beta_0 + \beta_1 MaraSalva_{i,t-1} + \beta_k X_{i,t} + \sum_{t=1}^{T-1} c_t D_t + \alpha_i + \lambda_{i,t}$$

- $Y_{i,t}$: Own-source revenue p.c., expenditure p.c., public service provision.
- $MaraSalva_{i,t-1}$: Homicides per 1,000 inhabitants attributed to MS-13 (**lagged one period**).
- $X_{i,t}$: Urban population, density, household income, energy consumption, capacity building.
- Specifications by municipality size: small (<10k), medium (10k–50k), large (>50k).

Spatial Model for Economic Activity

Cross-sectional spatial autoregressive model (GS2LS):

$$Y_i = \beta\, Gangs_i + \gamma\, Z_i + \theta\, W_{y,i} + \sigma\, W_{X,i} + \varepsilon_i$$
$$\varepsilon_i = \lambda\, W_{\varepsilon,i} + \mu_i$$

- Y_i: Electricity consumption (proxy for economic activity).
- W: Spatial weights matrix (shared border $= 1$).
- θ: Spillover of economic activity to neighbors.
- σ: Spillover of gang presence to neighbors' economic activity.
- λ: Spatial autocorrelation in unobservables.

Results

Effect of MS-13 on Own-Source Revenues

- Gang presence reduces own-source revenues per capita.
- Effect driven by **mid-size municipalities** (10k–50k pop.): $-\$5.6$ per homicide/1,000 inh.
- Not significant for small or large municipalities.

Effect of MS-13 violence on per capita own-source revenue

	(1) All municipalities	(2) Population < 10k	(3) 10k < Population < 50k	(4) Population > 50k
Maras	−1.136*	−0.426	−5.613*	−22.287
	(0.829)	(0.623)	(2.864)	(17.779)
Kw_mm	0.217	0.133*	0.853*	0.148**
	(0.125)	(0.179)	(0.435)	(0.054)
UrbanPopul	−0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Transf_pc	−0.130*	−0.015	−0.095	0.740
	(0.049)	(0.012)	(0.092)	(0.627)
Density	−0.016	−0.056	0.245	6.105
	(0.129)	(0.161)	(0.982)	(7.532)
Garbage_sv	2.204	0.451	10.871***	17.144
	(1.971)	(1.341)	(3.853)	(19.793)
year = 2012	2.151	−1.163	1.669	−3.070
	(1.277)	(2.761)	(1.841)	(3.724)
year = 2013	3.877*	−0.972	1.200	0.012
	(1.563)	(2.986)	(1.455)	(4.234)
year = 2014	6.029***	0.103	4.486	−1.869
	(1.800)	(3.136)	(2.769)	(5.912)
year = 2015	8.008***	1.394	6.095	0.774
	(1.979)	(3.065)	(3.203)	(6.177)
year = 2016	10.145***	1.122	9.722*	3.307
	(2.398)	(3.072)	(3.891)	(7.078)
year = 2017	10.681***	2.007	9.856**	4.288
	(2.202)	(3.014)	(3.568)	(6.013)
year = 2018	13.265***	3.964	10.250**	9.977
	(2.655)	(3.241)	(3.588)	(8.303)
Constant	27.078***	15.305***	9.079	−5.850
	(5.487)	(3.172)	(7.049)	(26.326)
Observations	1308	465	646	197

Standard errors in parentheses.
Robust standard errors in parentheses.
*$p < 0.05$.
**$p < 0.01$.
***$p < 0.001$.

Effect of MS-13 on Expenditures

- Significant negative impact on municipal spending per capita.
- For each additional homicide/1,000 inh., expenditure drops by \$17.9.
- Largest impact in **small municipalities**: $-$\$20 per homicide/1,000 inh.

Effect of MS-13 on expenditures

	All municipalities	Population < 10k	10k < Population < 50k	Population > 50k
Maras	−17.694***	−20.112**	−0.489	−38.447
	(6.347)	(12.785)	(7.180)	(29.544)
Kw	−0.006	−1.788*	0.137	−0.033
	(0.084)	(1.590)	(0.00)	(0.020)
UrbanPopul	−0.000***	0.001	−0.000	−0.000**
	(0.000)	(0.001)	(0.000)	(0.000)
Total_revenue_pc	1.077***	2.454***	0.891***	0.675***
	(0.210)	(0.611)	(0.218)	(0.184)
Density	8.661*	7.798*	−4.102	−24.177
	(2.687)	(2.425)	(2.848)	(16.557)
year = 2012	−6.980	−10.246	−6.819	6.558
	(3.843)	(9.701)	(3.643)	(3.620)
year = 2013	−5.096	−10.290	−2.824	5.885
	(4.470)	(10.244)	(4.646)	(4.177)
year = 2014	15.228**	18.975	12.621**	11.848
	(4.676)	(12.534)	(4.333)	(7.225)
year = 2015	−1.229	−12.911	0.766	1.515
	(3.937)	(10.887)	(4.687)	(3.756)
year = 2016	−14.906***	−37.984***	−8.827	−0.054
	(3.937)	(11.006)	(4.741)	(6.630)
year = 2017	−2.615	−20.951	2.251	1.650
	(4.001)	(13.647)	(4.994)	(5.471)
year = 2018	−14.213**	−46.045***	−2.254	−7.461
	(4.597)	(12.693)	(5.082)	(6.557)
Constant	85.488***	50.137	84.611***	35.412***
	(9.396)	(29.655)	(5.641)	(4.288)
Observations	1331	479	655	197

Standard errors in parentheses.
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Current vs. Capital Expenditure

- Capital expenditure hit hardest: $-\$19.7$ per homicide/1,000 inh.
- Current expenditure: $-\$2.9$ per homicide/1,000 inh.
- Effects concentrated in **small municipalities** ($<$10k pop.).

Effect of MS-13 on current & capital expenditure by size

	All municipalities	Population < 10k	10k < Population < 50k	Population > 50k	All municipalities	Population < 10k	10k < Population < 50k	Population > 50k
Maras	−2.866**	−3.884**	2.426	−15.151	−19.702***	−16.949*	−5.381	−9.386
	(1.058)	(1.193)	(2.328)	(13.248)	(4.639)	(7.578)	(7.044)	(26.668)
Kw	0.076***	−0.587	0.133***	0.008	−0.060	−1.388	−0.111	−0.028
	(0.017)	(0.324)	(0.036)	(0.013)	(0.058)	(0.940)	(0.088)	(0.024)
Urban_popul	−0.000***	−0.000	0.000	−0.000	−0.000***	0.001	−0.001***	−0.000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)
Transf_pc	0.557***	0.367***	0.526***	0.764***	0.352***	1.884***	0.133	0.429***
	(0.025)	(0.041)	(0.040)	(0.036)	(0.072)	(0.189)	(0.068)	(0.071)
Density	0.625**	0.580*	−0.315	−6.328	6.150***	7.117***	−2.580	−13.676
	(0.230)	(0.237)	(0.829)	(5.975)	(1.032)	(1.470)	(2.560)	(12.242)
year = 2012	3.472***	3.617**	3.831***	2.436	−9.808**	−13.757	−10.330**	4.255
	(0.840)	(1.304)	(1.151)	(2.316)	(3.686)	(8.657)	(3.649)	(4.766)
year = 2013	6.006***	6.342***	6.253***	3.705	−9.851**	−16.397	−8.229*	2.579
	(0.865)	(1.382)	(1.180)	(2.328)	(3.778)	(8.988)	(3.672)	(4.790)
year = 2014	7.682***	10.061***	7.777***	3.264	9.239*	9.893	6.216	9.025
	(0.874)	(1.394)	(1.195)	(2.333)	(3.789)	(9.021)	(3.685)	(4.797)
year = 2015	7.099***	10.290***	7.174***	0.971	−6.083	−21.976*	−4.612	1.138
	(0.885)	(1.400)	(1.241)	(2.418)	(3.797)	(8.948)	(3.736)	(4.957)
year = 2016	5.153***	7.172***	4.629***	3.903	−16.436***	−43.127***	−10.737**	−3.314
	(0.933)	(1.478)	(1.363)	(2.680)	(3.958)	(9.303)	(3.958)	(5.462)
year = 2017	7.660***	9.818***	6.843***	6.867*	−6.916	−27.692**	−1.818	−4.680
	(0.976)	(1.618)	(1.418)	(2.799)	(4.137)	(10.065)	(4.101)	(5.696)
year = 2018	7.554***	9.008***	8.392***	2.947	−17.908***	−51.450***	−7.456	−9.421
	(1.010)	(1.679)	(1.391)	(2.683)	(4.218)	(10.409)	(3.938)	(5.464)
Constant	23.018***	35.489***	19.515***	13.220***	70.934***	25.025*	73.683***	24.461***
	(1.604)	(2.921)	(2.808)	(3.021)	(4.949)	(11.479)	(4.093)	(5.795)
sigma_u								
Observations	1331	479	655	197	1331	479	655	197

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Effect of MS-13 on Public Service Provision

- Garbage collection coverage drops by 0.2% per homicide/1,000 inh.
- Effect significant in **mid-size municipalities**.
- No significant impact for small or large municipalities.

Effect of MS-13 on public service provision

	All municipalities	Population < 10k	10k < Population < 50k	Population > 50k
Maras	−0.002*	0.004	−0.002*	0.000
	(0.001)	(0.006)	(0.001)	(0.000)
Urban_popul	0.000***	0.000	0.000**	0.000*
	(0.000)	(0.000)	(0.000)	(0.000)
Household_income	0.000***	0.000**	0.000***	−0.000
	(0.000)	(0.000)	(0.000)	(0.000)
PC_own_rev	0.000	−0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.001)
Density	0.001	0.000	0.002	−0.012
		(0.006)	(0.010)	
	(0.006)			(0.028)
MunCapacBuild	0.025	0.049*	−0.010	−0.032
	(0.015)	(0.023)	(0.020)	(0.020)
year = 2011	−0.014	−0.031*	−0.006	0.011
	(0.010)	(0.016)	(0.015)	(0.009)
year = 2012	0.002	0.023	−0.018	0.009
	(0.011)	(0.021)	(0.013)	(0.009)
year = 2013	0.006	0.029	−0.009	0.016
	(0.018)	(0.038)	(0.021)	(0.016)
year = 2014	0.016	0.029	0.006	0.026*
	(0.019)	(0.041)	(0.023)	(0.012)
year = 2015	0.024	0.046	0.004	0.020
	(0.019)	(0.038)	(0.023)	(0.013)
year = 2016	0.041*	0.089*	0.012	0.007
	(0.020)	(0.041)	(0.024)	(0.018)
year = 2017	0.025	0.072	−0.005	0.008
	(0.021)	(0.045)	(0.026)	(0.015)
year = 2018	0.034	0.086	0.006	0.017
	(0.023)	(0.047)	(0.026)	(0.014)
Constant	0.194***	0.205**	0.085*	0.517***
	(0.042)	(0.070)	(0.037)	(0.059)
Observations	1391	520	662	209

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Effect on Local Economic Activity & Spillovers

- Each MS-13 homicide reduces local electricity consumption by 3.1 million kWh.
- Spillover: consumption in **neighboring** municipalities drops by 5.2 million kWh.
- Average municipal consumption $\approx$ 29 million kWh $\Rightarrow$ $\approx$10% reduction.

Impact of MS-13 on economic activity

	kWh
MaraSalva	−3.107**
	(0.996)
luminosity	0.013***
	(0.001)
PobUrb	0.000
	(0.000)
ingfa	0.059**
	(0.020)
gasto_pc	−0.005
	(0.011)
agedpopulation	143861
	(87197.9)
Constant	−51.824***
	(13.071)
Wma	
MaraSalva	−5.250*
	(2.149)
kw_mm	0.074
	(0.081)
All_gangs	
Observations	144

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Robustness

- **Other gangs**: Results similar when including homicides from gangs other than MS-13.
- **2SLS / IV estimation**:
  - Own-source revenues: MS-13 has significant negative effect (log specification).
  - Expenditures: All gangs combined show significant negative effect.
- Results consistent across multiple specifications.

Discussion & Policy Implications

Discussion

- Gangs hollow out municipal governance:
  - Lower revenues $\Rightarrow$ greater dependence on central transfers.
  - Lower spending $\Rightarrow$ reduced service provision.
  - Spillovers $\Rightarrow$ damage extends beyond gang-controlled areas.

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- Relationship is **parasitic**, not symbiotic:
  - No drug trafficking ⇒ no incentive to provide order.
  - Extortion competes directly with municipal tax collection.
- Effects **vary by municipality size**:
  - Mid-size: most vulnerable on revenues and services.
  - Small: most vulnerable on expenditures.
  - Large: more resilient (diversified economy, greater administrative capacity).

Policy Implications

Bukele's response: Recentralization + militarized security.

- Sharp cuts in FODES transfers.
- Central approval required for local investment.
- Municipalities reduced from 262 to 44 (2023).

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Alternative: Reconstruct decentralization.

- Modernize the municipal tax code ⇒ introduce **urban property tax** (less subject to extortion).
- Revise FODES formula ⇒ aid mid-size municipalities most vulnerable to extortion.
- Insulate Municipal Works Directorate from political pressure ⇒ technocratic criteria for investment.

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Don't abandon decentralization — **refine it** and exploit complementarities with the national government.

Thank You